Mini Project (Unit 2): Student Grading Automation System

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Aim

To design and implement a Python program that automates student grading — accepting marks for multiple students across several subjects, calculating each student's total, average, and letter grade, determining pass/fail status, and generating a final class report card along with the class topper.

Concepts Used

- **Functions** — to modularize grade calculation, result computation, data collection, and report display
- **Loops (for / while)** — to iterate over students and their subject-wise marks
- **Conditionals (if / elif / else)** — to decide grade boundaries and pass/fail status
- **Dictionaries and Lists** — to store each student's name mapped to their list of marks
- **Built-in functions** — sum(), all(), for aggregation and validation
- **String formatting (f-strings, alignment specifiers)** — to print a neatly aligned report card

Algorithm

1. Define the list of subjects (Maths, Physics, Python).
2. Ask the user for the number of students.
3. For each student, collect their name and marks in every subject, storing them in a dictionary.
4. For each student, calculate total = sum of marks and average = total / number of subjects.
5. Assign a grade based on the average: A+ (>=90), A (>=80), B (>=70), C (>=60), D (>=40), else F.
6. Mark the student as 'Pass' only if every individual subject mark is >= 35 and the grade is not F, otherwise 'Fail'.
7. Print a formatted report card listing every student's marks, total, average, grade, and result.
8. Track and display the class topper — the student with the highest average.

Source Code

```
# Mini Project (Unit 2): Student Grading Automation System
# Concepts used: functions, loops, conditionals, dictionaries, string formatting

SUBJECTS = ["Maths", "Physics", "Python"]


def calculate_grade(average):
    if average >= 90:
        return "A+"
    elif average >= 80:
        return "A"
    elif average >= 70:
        return "B"
    elif average >= 60:
        return "C"
    elif average >= 40:
        return "D"
    else:
        return "F"


def calculate_result(marks):
    total = sum(marks)
    average = total / len(marks)
    grade = calculate_grade(average)
    status = "Pass" if all(m >= 35 for m in marks) and grade != "F" else "Fail"
    return total, average, grade, status


def collect_students():
    students = {}
    n = int(input("Enter number of students: "))
    for i in range(n):
        print(f"\n--- Student {i + 1} ---")
        name = input("Enter student name: ").strip()
        marks = []
        for subject in SUBJECTS:
            mark = int(input(f"Enter marks in {subject} (out of 100): "))
            marks.append(mark)
        students[name] = marks
    return students


def display_report_card(students):
    print("\n" + "=" * 70)
    print(f"{'STUDENT GRADE REPORT':^70}")
    print("=" * 70)
    header = f"{'Name':<12}"
    for subject in SUBJECTS:
        header += f"{subject:>10}"
```

```
    header += f"{'Total':>8}{'Avg':>8}{'Grade':>8}{'Result':>10}"
    print(header)
    print("-" * 70)

    topper_name = None
    topper_avg = -1

    for name, marks in students.items():
        total, average, grade, status = calculate_result(marks)
        row = f"{name:<12}"
        for m in marks:
            row += f"{m:>10}"
        row += f"{total:>8}{average:>8.2f}{grade:>8}{status:>10}"
        print(row)

        if average > topper_avg:
            topper_avg = average
            topper_name = name

    print("=" * 70)
    print(f"Class Topper: {topper_name} (Average: {topper_avg:.2f})")
    print("=" * 70)


# ---- Program starts here ----
students = collect_students()
if students:
    display_report_card(students)
else:
    print("No student data entered.")
```

Output

Sample run — entering marks for 3 students (Rahul, Priya, Arjun) across Maths, Physics, and Python:

```
Mini Project - Student Grading System


$ python3 grading_system.py
Enter number of students: 3

--- Student 1 ---
Enter student name: Rahul
Enter marks in Maths (out of 100): 85
Enter marks in Physics (out of 100): 78
Enter marks in Python (out of 100): 92

--- Student 2 ---
Enter student name: Priya
Enter marks in Maths (out of 100): 95
Enter marks in Physics (out of 100): 91
Enter marks in Python (out of 100): 89

--- Student 3 ---
Enter student name: Arjun
Enter marks in Maths (out of 100): 45
Enter marks in Physics (out of 100): 38
Enter marks in Python (out of 100): 52

======================================================================
                        STUDENT GRADE REPORT
======================================================================
Name           Maths   Physics    Python   Total     Avg   Grade    Result
----------------------------------------------------------------------
Rahul             85        78        92     255   85.00       A      Pass
Priya             95        91        89     275   91.67      A+      Pass
Arjun             45        38        52     135   45.00       D      Pass
======================================================================
Class Topper: Priya (Average: 91.67)
======================================================================
```

Result

The Student Grading Automation program was executed successfully. It correctly computed the total, average, and letter grade for each student, applied the pass/fail rule, and identified Priya as the class topper with an average of 91.67 (Grade A+). The program can be extended with features like saving results to a file, computing subject-wise class averages, or exporting the report card to a spreadsheet.